Weizheng Wang

Computer Vision · Multimodal Learning · UAV/Edge AI

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Research Profile

Master of Artificial Intelligence and Machine Learning graduate researching **multimodal decision reliability, visual memory and efficient vision**. Co-first-author work uses protocol-conditioned evaluation, matched interventions, structured pruning and detector feature analysis to study how model behavior changes with evidence, task rules and resource budgets. Research-software practice includes reusable agent skills, workflow design, MindSpore integration and Jetson deployment.

Education

Adelaide University

Adelaide, Australia

Master of Artificial Intelligence and Machine Learning

Sep. 2025 – Oct. 2026

- Completed the two-year master's programme within one year.
- Completed 18 August 2026; degree conferred 7 October 2026.
- Recipient of the 2025 University of Adelaide Global Citizens International Scholarship (highest tier).

North China Institute of Science and Technology

Langfang, China

Bachelor's Degree in Computer Science and Technology

Sep. 2021 – Jun. 2025

- GPA: 4.02/5.
- Coursework and research focused on deep learning, computer vision, and UAV-based edge deployment.

Research and Engineering Experience

Zayed University

Remote

Research Assistant – MulRobBench: Safe Multimodal UAV Agents

Feb. 2026 – Aug. 2026

- Co-designed a decision-level study of whether UAV agents preserve mission rules under degraded observations and ambiguous instructions: 3,024 samples, 17 task-taxonomy nodes and 12 scoring dimensions.
- Designed semantic scoring and strict diagnostics to distinguish decision quality, output validity and safety-policy adherence under controlled observation and language conditions.
- Across 17 models, semantic, structural and action-safety rankings diverged. For two representative models on 20 matched anchors, modality removal changed 4–15 of 20 action selections per model, revealing input sensitivity.
- Contributed methodology, formal analysis, software and manuscript drafting; co-first author, manuscript submitted, arXiv:2607.23870.

Institute of Software, Chinese Academy of Sciences

Remote

Research Intern – Model Adaptation, Fine-tuning & Evaluation

Feb. 2025 – Jun. 2025

- Investigated cross-framework adaptation for forecasting, image classification and sentiment analysis, adapting Autoformer, BEiT and ALBERT fine-tuning workflows to MindSpore/MindNLP.
- Documented PyTorch/MindSpore comparisons on tourism_monthly, CIFAR-10 and IMDb, including Autoformer padding/operator compatibility; contributed fine-tuning scripts and reports through three merged pull requests. [Autoformer] [BEiT] [ALBERT]

Beijing Elite Intelligence Company & NCIST

Beijing & Langfang, China

Research Intern – Automatic Inspection and Rescue Based on Drone Nest

Jan. 2024 – Jul. 2024

- Addressed aerial truck monitoring and mine-inspection perception under edge-compute constraints, combining YOLOv8 and DeepSORT with standing-water counting and conveyor-belt deviation detection.
- Contributed detector optimization and ONNX deployment for real-time edge inference in long-duration UAV missions, linking perception methods to inspection-system requirements.

Hebei Province Key Sci. & Tech. R&D Project (No. 19270318D)

Hebei, China

Undergraduate Researcher – Baiyangdian Ecological IoT 3D Monitoring

Jun. 2023 – Jan. 2024

- Investigated efficient mining-truck detection with YOLOv7-tiny (Darknet); the existing evaluation reports a 2% recognition-accuracy gain with limited FPS loss.
- Evaluated infrared-visible YOLOv8 perception for low-altitude UAV monitoring; the existing laptop test reports 65% mAP at 30 FPS to inform edge-deployment choices.

Accepted Papers, Preprints & Submitted Manuscripts

†Equal contribution / co-first authors. *Corresponding author.

FHS-DEIM: Frequency-Aware Hybrid Encoding with Spiral Mamba for Infrared UAV Search and Rescue
2026 | Submitted to Journal of King Saud University Computer and Information Sciences | Technical checks passed

MulRobBench: A Decision-Level Benchmark for Safe and Security-Policy-Compliant Multimodal UAV Agents

Belal S. Alsinglawi†, **Weizheng Wang**†, Junyi Wu†, Lianhai Lin, Mérouane Debbah, Izzat Alsmadi

2026 | Manuscript submitted | arXiv:2607.23870

Co-designed a decision-level study of rule adherence under degraded UAV observations and ambiguous instructions: 3,024 samples, 17 task-taxonomy nodes and 12 scoring dimensions. Evaluation of 17 models exposed different rankings across semantic decision quality, structural validity and action safety. For two representative models on 20 matched anchors, modality removal changed 4–15 of 20 action selections per model; contributed methodology, formal analysis, software and drafting.

Hilbert–Haar Representation with Tri-Modal Multi-Dimensional Pruning for Maritime Human Search from Uncrewed Aerial Vehicle Imagery

Junyi Wu†, **Weizheng Wang**†, Liqin Tian, Qing Tian, Yong Niu, Yi Jiang

2026 | Submitted to IEEE JSTARS

Targets maritime small-object detection under compute constraints with Hilbert-Mamba context modeling, Haar wavelet downsampling and structured pruning across CNN, Transformer and state-space widths. H2T-DEIM-M achieves 51.1% AP on SeaDronesSee and 64.8% on AFO, with 32.1 FPS on Jetson Orin Nano 8GB under the manuscript's end-to-end protocol; contributed methodology, software and validation.

A Multi-Source Data Fusion and PSO-Transformer Decision Support System for Gold Price Volatility Forecasting and Risk Warning

Xueping Diao*, Xueqiang Lv, **Weizheng Wang**, Junyi Wu

2026 | Mathematics (MDPI) | Accepted for publication, 2 November 2026

Combines structured market data with TFN-based video sentiment and Qwen-2.5 news sentiment, using PSO-tuned Transformer forecasting and three-tier risk warnings. Reports 74.5% risk-tier classification accuracy in a 102-trading-day backtest.

TRIM: Dual-Budget Structured Pruning with Information-Modulated Saliency for CNN, Transformer, and Mamba Vision Models

Weizheng Wang†, Junyi Wu†, Qing Tian, Liqin Tian

2026 | Submitted to IEEE ICASSP 2027 | [Repository]

Addresses mismatches between analytical pruning budgets and measured model costs using dependency-closed units, information-modulated Taylor/Fisher saliency, coupled MILP allocation and measured-budget repair. On ImageNet-1K, Top-1 losses are 0.52/0.66/0.58 percentage points for ResNet-50/DeiT-Base/VMamba-V2 Tiny, retaining at most 69.5% of parameters and 69.4% of forward GFLOPs with zero measured budget violation.

When Retrieval Is Not Binding: A Diagnostic Benchmark for Visual Episodic Memory

Weizheng Wang†, Junyi Wu†, Liqin Tian, Qing Tian

2026 | Submitted to IEEE ICASSP 2027 | [Repository]

Designed matched memory-off, wrong-entity and shuffled-binding interventions while fixing queries, candidates, features and memory envelopes. The development cohort contains 48 objects, 384 observations and 192 queries; on 128 memory-visible queries, three of four training-free readers are unchanged by shuffled binding, separating content benefit from episodic binding sensitivity.

ManiSeg: A Semantic-Entropy Driven Manifold Collision Framework for High-Dimensional Feature Evaluation of Detectors

Junyi Wu†, Zitong Shen†, **Weizheng Wang**†, Xiaoan Shi, Yi Jiang, Liqin Tian

2026 | Submitted to IEEE ICASSP 2027 | [Code]

Co-developed semantic-entropy-driven granulation, geometric manifold-collision-based adaptive three-way decision and a composite Manifold-Semantic Fidelity metric; evaluated detector feature representations on COCO across multiple model variants.

EmergUAV-Bench: Protocol-Conditioned Evaluation of Vision-Language-Action Models for Emergency UAV Decisions

Junyi Wu†, **Weizheng Wang**†, Yi Jiang†, Liqin Tian

2026 | Submitted to IEEE ICASSP 2027 | [Code]

Co-developed a 1,280-sample benchmark combining RGB, LiDAR and pose observations with six language conditions, six core emergency tasks and 16 decision facets. Evaluated 13 multimodal models; the best protocol score is 0.4165, with distinct leaders in protocol agreement, safe-action accuracy and unsafe-action avoidance.

Peer-Reviewed Publications

A Lightweight Thermal Denoising and Occlusion-Robust Infrared Detection Model for Substation Equipment — *IEEE ICIP, pp. 1–6*

Junyi Wu, **Weizheng Wang**, Yong Niu, Xiaoan Shi, Liqin Tian — *2026*

- PO-YOLO achieved 83.5% mAP50 and 56.9% mAP50:95 on ISED-1K, improving the YOLOv11 baseline by 3.4 and 2.0 points while reducing computation by 6.35%. [Paper] [Code]

A lightweight insulator defect detection algorithm based on drone images for power line inspection — *Engineering Research Express 7, 045355*

Junyi Wu, **Weizheng Wang**†, Wanyan Du†, Zhi Liu, Lianhai Lin, Yuchen Cui, Chong Wang, Liqin Tian — *2025*

- LFL-YOLO achieved 97.4% accuracy while reducing parameters by 23.5% and computation by 9.6% versus YOLOv8n. †Equal contribution. [Paper]

Real-time monitoring of trucks used in open pit based on aerial video of UAV — *IEEE RAIIC, pp. 130–138*

Weizheng Wang, Junyi Wu, Yongfei Li — *2024*

- Developed a real-time UAV-based monitoring system for open-pit mine trucks using aerial video. [Paper]

Research on industrial meter readings based on improved YOLOv8 — *Proc. SPIE 13181, 131815G*

Junyi Wu, **Weizheng Wang**, Lei Yang, Xuesong Ni, Liqin Tian — *2024*

- Proposed an improved YOLOv8 approach for robust industrial meter reading. [Paper]

Honors & Leadership

2025	**Recipient,** University of Adelaide Global Citizens International Scholarship (highest tier)	*Australia*
2024	**Principal Investigator,** National Undergraduate Innovation Training Program (202411104021)	*China*
2023	**Grand Award,** National 3D Digitization Innovation Competition	*China*
2023	**Second Prize,** Challenge Cup Science and Technology Invention and Creation Track	*China*
2023	**First Prize,** National College Students' Data Analysis Competition	*China*

Patents

A Dual-Optical Personnel Search and Rescue Positioning Visual System on Drones — *China National Intellectual Property Administration*

Co-inventor — *Nov. 2024 - Present*

- Invention patent under substantive examination. Publication No. CN 119429233 A.

Automatic Battery Change Arm for Drones — *China National Intellectual Property Administration*

Co-inventor — *Dec. 2023*

- Utility model patent granted. Patent No. ZL 2023 2 3426097.6.

Agent Workflows & Reusable Skills

APEX Harness & Academic Research Harness

System design and research-software practice

[APEX] *[Academic Research Harness]*

Designed modular skills for evidence-guided agent work: semantic-risk tiers, verification and stopping rules in APEX; task-based reference loading and claim-preserving manuscript workflows in Academic Research Harness. Implemented package validation, project initialization, literature-index checks and reading-trace tooling; designed seven synthetic writing cases with independent output review to examine reference selection and scientific fidelity.

Research Projects

National Undergraduate Innovation and Entrepreneurship Training

Project lead: 202411104021 (2024); participant in collaborating student projects

Collaborating projects: 202511104021, 202511104025, 202611104287

Undergraduate research in UAV perception, automated inspection and edge inference. Collaborating projects are led by Yuxia Gao, Lucan Tang and Yi Jiang, respectively.

National Key R&D: Multimodal Perception and Emergency Decision Research

Research participant | Maritime detection, model compression, visual memory and UAV decision evaluation

2024YFC3016800; 2024YFC3016805

Studied efficient maritime perception, representation quality, episodic binding and protocol-conditioned UAV decisions through H2T-DEIM, TRIM, ManiSeg, the visual-memory diagnostic and EmergUAV-Bench; contributed model implementation and comparative evaluation.

Intelligent Mine Inspection and Monitoring

Research participant | UAV video analytics and industrial inspection

2022YFC3004702-4; HNCTR23-05

Participated in research associated with coal-mine disaster monitoring and digital decision systems, and long-distance coal-flow roller/belt monitoring. Related undergraduate work includes mining-truck detection/tracking, standing-water counting and conveyor-belt deviation detection.

Multimodal Airborne Search-and-Rescue Equipment

Research participant | Equipment development and demonstration | Project budget: RMB 800,000

Related regional programs: 202622003010002; 25350802D; 2025011099

Related work includes infrared-visible UAV detection, personnel search/localization and edge inference. Regional participation covers Hebei major science and technology support, Hebei innovation-capacity enhancement and Langfang science and technology support programs.

IoT Research Platforms and Collaborative Research Funds

Research participant | Sensing, visual detection and research software

21567693H; 2017-ZJ-Y21; 3142024038; 3142026024; 262Y5403D

Hebei IoT Monitoring Technology Innovation Center; Qinghai IoT Key Laboratory; Fundamental Research Funds for the Central Universities; Hebei collaborative innovation special project on livelihood security and social safety. The undergraduate Baiyangdian project (19270318D) is detailed in Research Experience.

Selected Open-Source Projects

- **Jetson VLM Lab**: Research tooling for resource-constrained GGUF vision-language inference on Jetson, with llama.cpp/Docker deployment and text/image smoke evaluation.
- **DEIM Jetson Acceleration**: Detector-deployment evaluation of FP16/INT8 correctness, latency and power through ONNX/TensorRT, supporting precision/resource trade-off analysis.
- **Video-based Intelligent Alert System**: Undergraduate thesis platform connecting multi-stream monitoring, open-vocabulary perception, alerts and multimodal Q&A.
- **DailyPaper**: Automated AI paper collection with a searchable, categorized web archive.
- **CleanSlateTab**: Browser new-tab extension with configurable search, shortcuts, wallpapers and WebDAV sync.
- **SnapPin**: C++ Windows utility for screenshots, annotation, OCR and pinned images.

Technical Skills

Inference & Edge	TensorRT, ONNX, Jetson, llama.cpp/GGUF; FP16 / selective Q-DQ INT8 comparisons, output correctness, synchronized latency and power measurements; Docker, Linux, CUDA runtime
DL & Vision	PyTorch, MindSpore/MindNLP, Transformers; YOLO/DEIM, OpenCV/DeepSORT, detection/tracking, structured pruning; controlled perturbations and matched diagnostics; multimodal evaluation, visual memory, thermal-visible perception, feature analysis and granular computing
Systems & Tools	Python, PowerShell, SQL; Django/Flask, SQLite/MySQL, ZLMediaKit/FFmpeg; Git, CMake; C++, JavaScript/TypeScript, Java, LaTeX; browser extensions and GitHub Pages